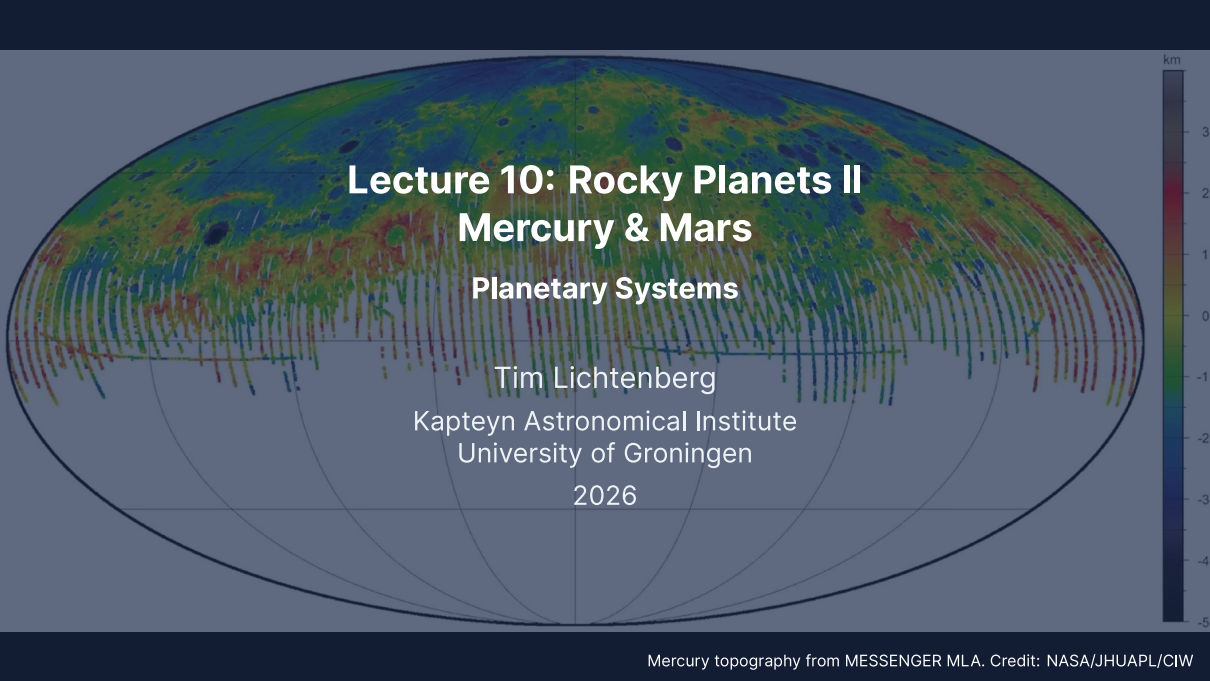


A global map of Mercury's topography in a pseudocylindrical projection. The map uses a color scale from dark blue (-5 km) to dark red (3 km). The equatorial region is predominantly red and orange, indicating higher elevations, while the polar regions are mostly blue and green, indicating lower elevations. Numerous impact craters of various sizes are visible across the surface. A vertical color bar on the right side of the map provides the elevation scale in kilometers.

Lecture 10: Rocky Planets II

Mercury & Mars

Planetary Systems

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2026

Learning Objectives

By the end of this lecture, you will be able to:

- Explain Mercury's anomalously large iron core and the spin-orbit resonance
- Describe Mars' interior, surface, and the InSight seismic results
- Derive the Jeans escape flux and apply it to atomic H, H_2 , and CO_2 on Mars
- Distinguish thermal from non-thermal atmospheric escape mechanisms
- Compare Mercury and Mars as opposite limits of terrestrial-planet evolution
- Outline the missions returning data from both worlds in the 2020s and 2030s

Recap: Lecture 9

- Earth and Venus: nearly identical bulk parameters, drastically different evolution
- Simpson–Nakajima limit: maximum OLR of a saturated water atmosphere $\sim 280 \text{ W m}^{-2}$
- Venus inside the runaway-greenhouse threshold; D/H $\sim 150\times$ Earth's
- Plate tectonics + magnetic field + biosphere = the Earth-only four-way coincidence

Today: the smaller siblings, Mercury and Mars. Two opposite limits: largest iron-core fraction (Mercury) vs. near-extinct dynamo (Mars). **Why both?**



1. Mercury Overview and Surface

Caloris Basin, Mercury. MESSENGER (NASA/JHUAPL)

Mercury at a Glance

Parameter	Value	Note
Mass	$0.0553 M_{\oplus}$	smallest planet
Radius (mean)	$0.383 R_{\oplus}$	2440 km
Mean density	5429 kg m^{-3}	uncompressed $\sim 5300 \text{ kg m}^{-3}$
Semimajor axis	0.387 AU	innermost
Orbital period	87.97 d	
Rotation period	58.65 d	3:2 spin-orbit resonance
Eccentricity	0.206	largest of any planet
T_{surf} (day/night)	700/100 K	no atmosphere to redistribute
Magnetic field	$\sim 1\%$ Earth's	weak but global

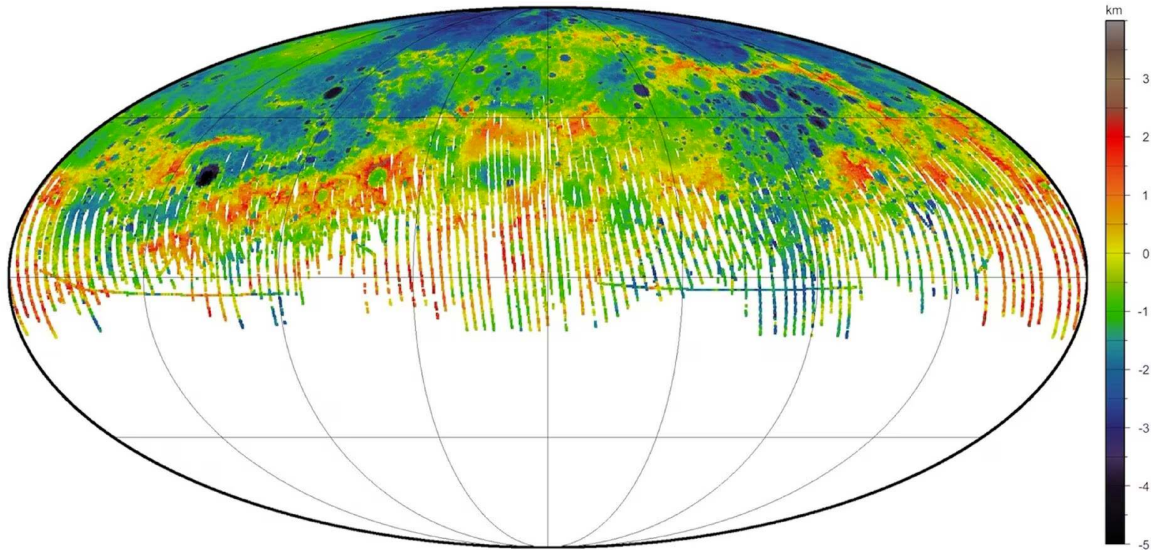
Source: NASA Mercury Fact Sheet (NSSDC)

The 3:2 Spin-Orbit Resonance

Mercury rotates exactly 3 times for every 2 orbits.

- Discovered by radar, Pettengill & Dyce (1965)
- Solar tidal torque on a permanent quadrupole moment (a bulge along the long axis) locks Mercury into the resonance
- Stable because of high orbital eccentricity ($e = 0.206$)
- Solar day on Mercury: 176 Earth days = 2 orbits
- Hot pole and cold pole alternate with orbital phase

Mercury is the only known planet in a non-1:1 spin-orbit resonance. The capture is dynamically robust because of eccentricity-driven tidal evolution.



MESSENGER MLA global topography. Credit: NASA/JHUAPL/CIW

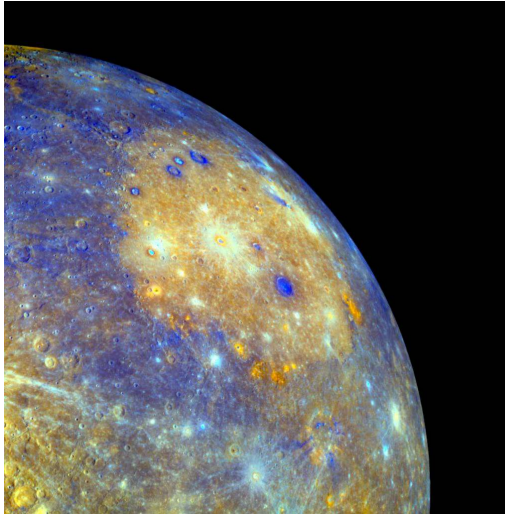
Reading Mercury's Topography

MESSENGER's Mercury Laser Altimeter mapped the planet at ~ 250 m vertical resolution.

- Total topographic range: ~ 10 km (less than Mars or the Moon)
- Highlands: ancient cratered terrain, > 4 Gyr old
- Smooth plains: volcanic infill, ~ 3.7–3.9 Ga
- **Caloris Basin** (~ 1550 km diameter): one of the largest impacts in the inner solar system
- **Lobate scarps**: thrust faults (crust overlapping under compression) from global contraction

Two geological epochs: pre-tectonic + heavy bombardment, then a long stagnant cooling.

Caloris Basin: A Mercury-Defining Impact



Credit: NASA/JHUAPL (MESSENGER)

- Diameter ~ 1550 km, formed ~ 3.8 Ga
- Antipodal "weird terrain" from focused seismic shockwave
- Concentric ring structure preserved
- Floor flooded by basaltic plains
- Major chronostratigraphic reference for Mercury

Lobate Scarps: Mercury Has Shrunk

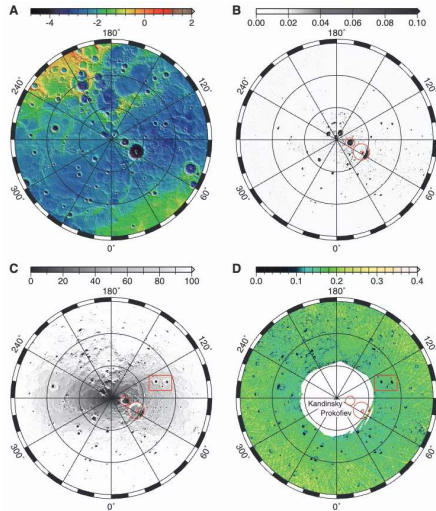


Credit: NASA/JHUAPL (MESSENGER)

- Mercury contracted as it cooled: total radial shrinkage ~ 7 km
- Compressional thrust faults form lobate scarps
- Scarp height up to 3 km, length > 1000 km
- Sets a constraint on heat-flow history

Source: Byrne et al. (2014)

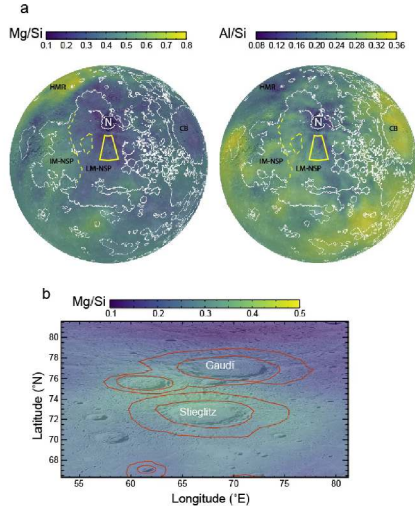
Polar Ice on the Hottest Planet



Credit: Neumann et al. (2013)

- Permanently shadowed polar craters never see sunlight
- $T < 100 \text{ K} \Rightarrow$ cold trap for water ice
- Panels A–D: MLA topography, reflectance, biannual-average insolation, max-illumination T all coincident
- Identity = H_2O ice (neutron spectrometer, Lawrence 2013); source: comet/asteroid impacts

Mercury's Surface Chemistry



Credit: Nittler et al. (2020)

Surprises from MESSENGER X-ray and gamma-ray spectrometers:

- High volatile abundances (S, K, Cl, Na) despite 700 K daytime
- Low FeO (< 2%): mantle is highly reducing
- Mg/Si ratio higher than Earth's
- Constrains formation under low oxygen fugacity (effective O_2 partial pressure)

Exosphere and Magnetosphere

Mercury has no real atmosphere; instead, a tenuous **exosphere** of atoms knocked off the surface.

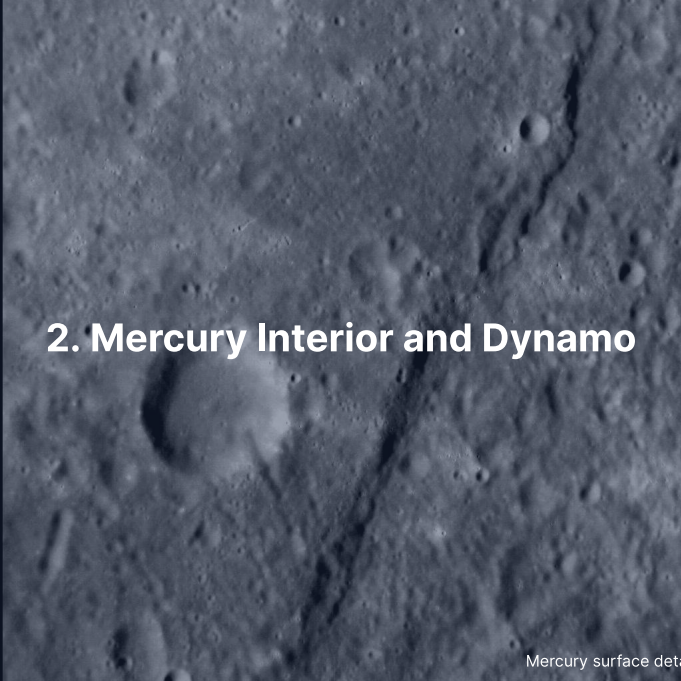
- Surface pressure $\sim 10^{-12}$ bar (effectively vacuum)
- Species detected: Na, K, Ca, Mg, He, H
- Sources: micrometeorite impact, photon-stimulated desorption, sputtering, thermal release
- Sodium tail extends $\sim 10^6$ km anti-sunward

Magnetosphere: small, asymmetric, driven by the planet's weak global dipole + solar-wind interaction.

Missions to Mercury

- **Mariner 10** (1974–1975): three flybys, ~ 45% of surface imaged
- **MESSENGER** (2011–2015): orbiter, full surface mapping, MLA, MAG, X/GRS
- **BepiColombo** (ESA/JAXA, launched 2018): two orbiters, Mercury orbit insertion late 2026
 - MPO: surface, interior, exosphere
 - Mio (MMO): magnetosphere, plasma environment

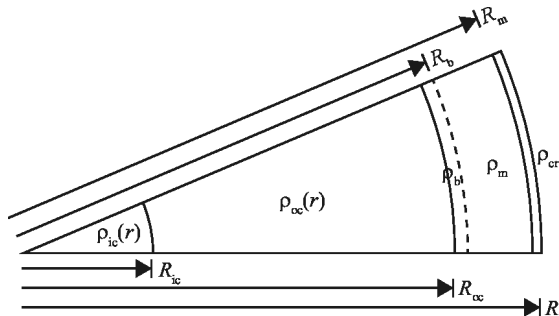
In a year and a half, BepiColombo's two orbiters will revolutionise our view of Mercury's interior, magnetosphere, and surface composition.

A grayscale image of the Mercury surface, showing a dense field of craters of various sizes. A prominent, dark, linear feature, likely a fault or a deep crater, runs diagonally from the upper right towards the lower center. The surface texture is highly irregular due to the numerous impact craters.

2. Mercury Interior and Dynamo

Mercury surface detail. MESSENGER (NASA/JHUAPL)

A Giant Iron Core

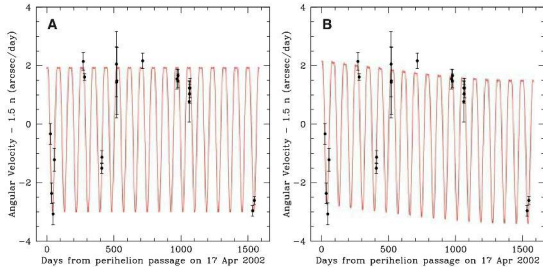


Credit: Margot et al. (2018)

Margot et al. libration measurements pin down the layers:

- Crust: ~ 35 km
- Mantle (silicate): ~ 400 km
- Outer core (liquid): below $\sim 2440 - 400 - 35 = 2005$ km radius
- Inner core (solid, possibly Fe-S)
- Core radius ~ 2020 km, $\sim 83\%$ of planetary radius
- Core mass fraction $\sim 74\%$

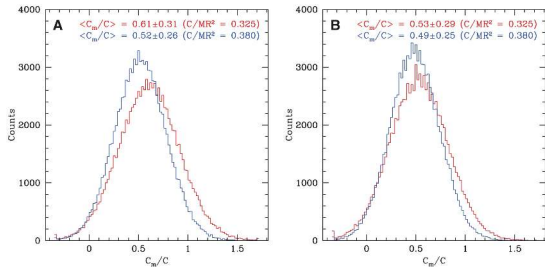
How We Know: Forced Libration



Credit: Margot et al. (2007), Fig. 3

- Mercury rocks slightly back and forth as it orbits
- Points: spin rate measured at each radar epoch; red curve: the fitted 88-day libration
- Libration amplitude depends on whether the mantle is decoupled from the core
- Earth-based radar speckle interferometry: amplitude 35.8 ± 2 arcseconds
- Implies a liquid outer core: only the silicate mantle librates freely

The Libration Measurement



Credit: Margot et al. (2007), Fig. 4

- Posterior on C_m/C (mantle moment / total moment) from Monte Carlo realisations
- Peaks at ~ 0.5 , inconsistent with a fully solid Mercury (which would peak near 1)
- Direct evidence the silicate mantle floats on a fluid core
- Combined with C_{20} and C_{22} from gravity, fully constrains C/MR^2 and core radius

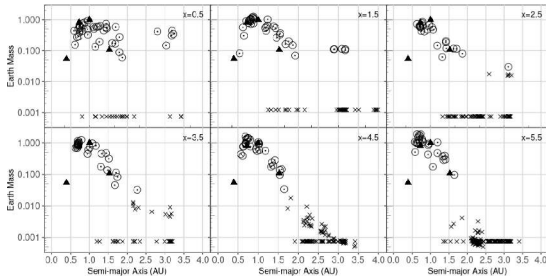
Why is Mercury so Iron-Rich?

Three competing hypotheses for Mercury's anomalous core fraction:

1. **One or more giant impacts** (leading hypothesis; Benz, Slattery & Cameron 1988; Chau et al. 2018): late hit-and-run collisions strip a silicate mantle from a proto-Mercury
2. **Vapour and aerosol stripping by an early luminous Sun:** thermal evaporation / wind-driven loss of part of an originally silicate-rich proto-Mercury
3. **Selective condensation in the inner nebula:** hot inner-disk midplane condenses Fe-Ni metal preferentially over silicates (no longer favoured)

MESSENGER surface volatiles (Na, S, K) disfavour thermal stripping; the giant-impact family is the modern leading candidate.

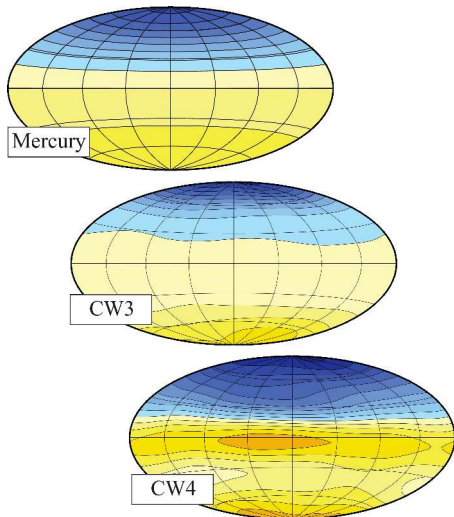
Impact-Stripping Outcomes



Credit: Franco et al. (2022), Fig. A1

- N-body (gravity-only) terrestrial-formation simulations: remnant mass vs. semi-major axis
- Six panels: surface-density slope $x = 0.5$ – 5.5 ; triangles = real planets
- Mercury-mass + iron-rich at the right a occurs in $\ll 1\%$ of trials
- Single-impact origin is dynamically rare

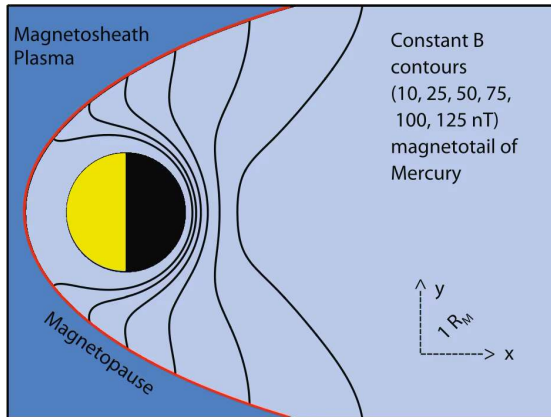
A Weak, Offset Dynamo



Credit: Wicht & Heyner (2017)

- Mercury's dipole offset
~ $0.2 R_{Hg}$ northward
- Surface field strength:
~ 200–400 nT (Earth's:
~ 50,000 nT)
- Dynamo (self-sustained field from convection) in a thin outer-core shell
- Stably stratified upper liquid-core layer suppresses high-degree harmonics; inner-core freezing asymmetry produces the offset

Mercury's Magnetosphere



Credit: Wicht & Heyner (2017)

- Smallest planetary magnetosphere in the solar system
- Subsolar magnetopause (boundary deflecting the solar wind): $\sim 1.5 R_{Hg}$
- Reconnection (magnetic field-line merging) events drive intense flux-transfer events
- BepiColombo will measure plasma + field structure

A grainy, black and white photograph of the Martian surface, showing numerous craters of various sizes and dark, irregular terrain. The image is framed by a film strip border.

3. Mars Overview and Moons

Mariner 4 view of Mars (1965). NASA/JPL

Mars at a Glance

Parameter	Value	Note
Mass	$0.107 M_{\oplus}$	one tenth of Earth
Radius (mean)	$0.532 R_{\oplus}$	3389 km
Mean density	3934 kg m^{-3}	lowest of the rocky planets
Semimajor axis	1.524 AU	
Orbital period	686.97 d	
Rotation period	24.62 h	similar to Earth
Axial tilt	25.2°	seasons like Earth's
Eccentricity	0.0934	strong climate forcing
T_{surf} (day/seasonal)	-130 to $+30^{\circ}\text{C}$	
Surface pressure	$\sim 6 \text{ mbar}$	0.6% of Earth's
Magnetic field	none global	local crustal anomalies

Source: NASA Mars Fact Sheet (NSSDC)

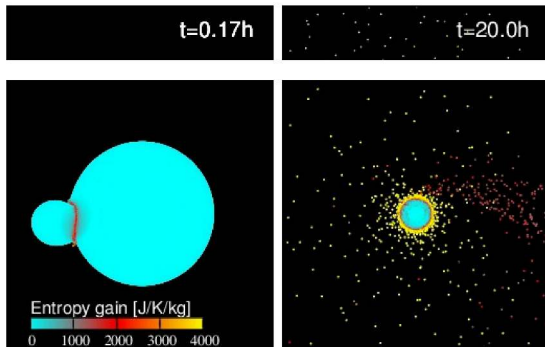
Phobos and Deimos

Mars' two tiny moons:

- **Phobos:** 22 km mean diameter, orbit at 9376 km, period 7.7 h (faster than Mars rotation)
- **Deimos:** 12 km, orbit at 23463 km, period 30.3 h
- Both have very low density ($\sim 1900 \text{ kg m}^{-3}$), consistent with rubble-pile internal structure
- Two competing origin scenarios:
 - Captured asteroids
 - In situ formation from a giant-impact debris disk

Phobos is below synchronous orbit and will tidally decay onto Mars in $\sim 30\text{--}50$ Myr.

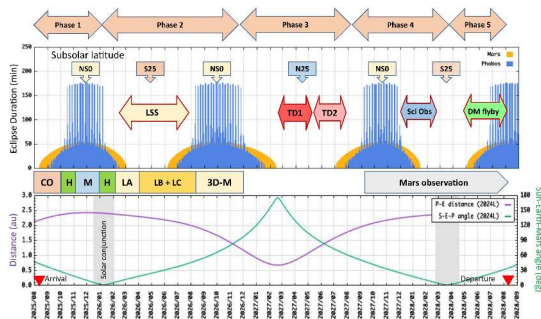
The Giant-Impact Origin Hypothesis



Credit: Hyodo et al. (2017)

- SPH simulation of a Borealis-scale impact
- Forms an extended debris disk around Mars
- Phobos and Deimos accrete from the disk
- Reproduces orbital constraints; composition still debated (predicted Mars-mantle vs. observed D-type)
- Predicts a fraction of Mars-derived material in the moons

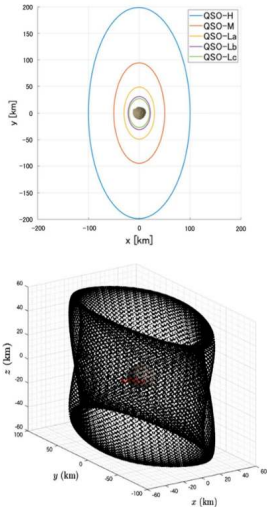
MMX: Sample Return from Phobos



Credit: Kuramoto et al. (2022), Fig. 4

- JAXA's Martian Moons eXploration: launch 2026, sample return 2031
- Operation plan for the three-year stay, in five phases
- **Top:** eclipses by Mars (orange) and Phobos (blue); arrows mark the touchdowns and the Deimos flyby
- **Bottom:** grey bands are solar conjunctions, radio link stops
- Will distinguish capture vs. impact origin

MMX Mission Architecture



- Quasi-satellite orbits (stable relative orbits) around Phobos
- **Top:** five planar orbits, QSO-H at ~ 200 km down to QSO-Lc at tens of km
- **Bottom:** a 3D orbit inclined 45°, to cover the whole surface
- Sampling phase: ~ 10 g of regolith (loose surface material)

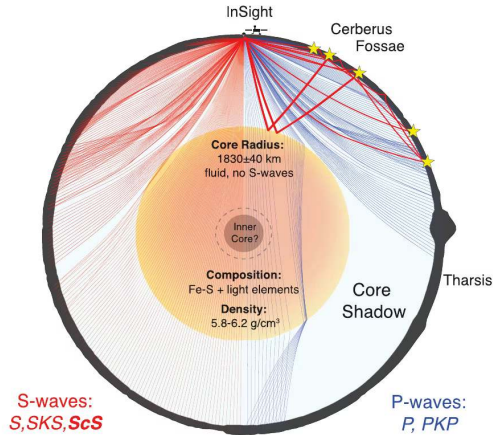
Credit: Kuramoto et al. (2022), Fig. 3

The image displays four terrestrial planets in a row, scaled to show their relative sizes. From left to right: Mercury is a small, grey, cratered sphere; Venus is a large, yellowish-brown sphere with a hazy atmosphere; Earth is a large sphere with blue oceans, white clouds, and brown/green landmasses; and Mars is a medium-sized, reddish-brown sphere with darker patches. The planets are set against a dark blue background.

4. Mars' Interior: The InSight Revolution

Terrestrial planets to scale. Credit: NASA/JPL-Caltech

Mars' Interior, Resolved

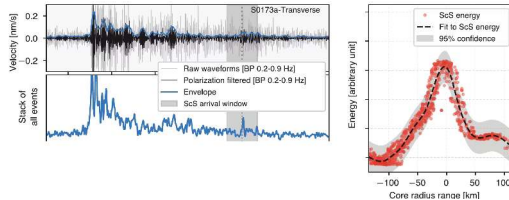


InSight SEIS results:

- Crust: 24–72 km thick, thinner in the north, thicker in the south
- Mantle to ~ 1560 km depth
- Tentative olivine–wadsleyite transition ~ 1000–1100 km (model-dependent)
- **Liquid iron-alloy core** of radius ~ 1830 km
- Core density ~ 6000 kg m^{-3} (Earth: ~ 10,900)

Credit: Stähler / Khan / Knapmeyer-Endrun (2021)

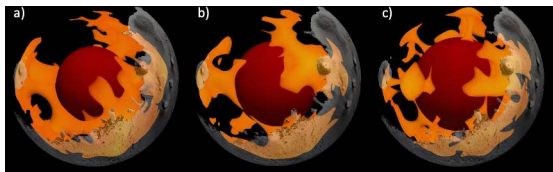
Reading a Marsquake Waveform



Credit: Stähler et al. (2021)

- P-wave arrival, then S-wave arrival
- Core reflections (ScS) detected: core radius ~ 1830 km, outer core liquid
- ~ 1300 events catalogued during the mission
- Largest event: S1222a, $M = 4.7$, May 2022

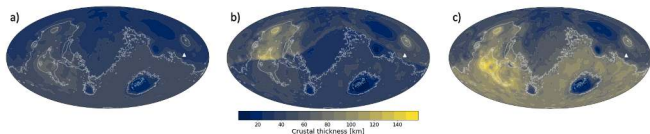
Mars Mantle Convection Today



Credit: Pleša et al. (2022)

- 3D thermal-evolution model consistent with InSight constraints
- A small number of long-wavelength upwellings (low-degree convection)
- Sluggish lower-mantle flow
- Tharsis upwelling persists to the present

Mars Crustal Thickness Map



Credit: Pleša et al. (2022), Fig. 1a-c

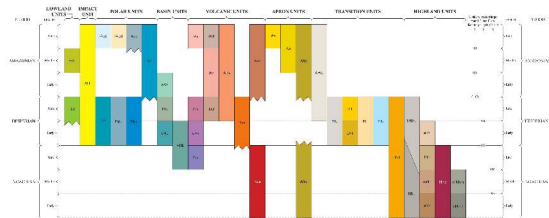
- All anchored to the seismic crust under InSight (triangle)
- **a**: mean 40.6 km, $\rho = 2550$
- **b**: mean 43.1 km, lowlands 3000, highlands 2600
- **c**: mean 71.4 km, $\rho = 3000 \text{ kg m}^{-3}$
- Dichotomy clear in **c**, near-absent in **b**: it depends on the assumed density



5. Mars Geology and Surface Highlights

Olympus Mons. NASA/JPL/Mars Express (ESA)

Mars' Geological Periods

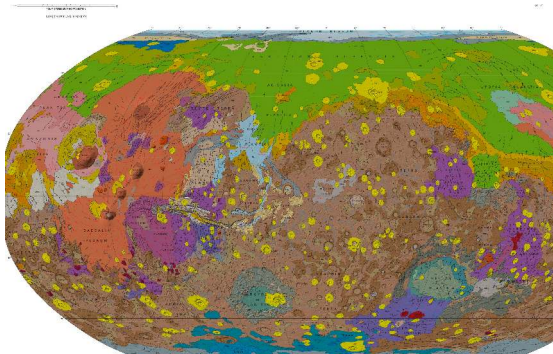


Credit: Tanaka et al. (2014)

Three classical periods:

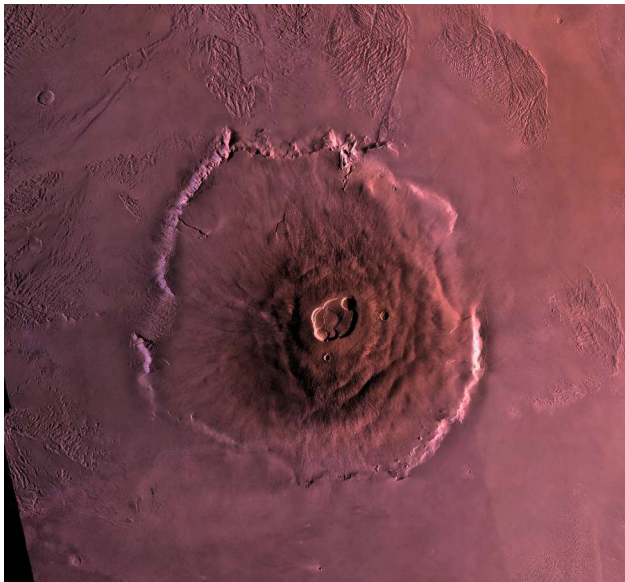
- **Noachian** (> 3.7 Ga): heavy bombardment, valley networks, possible warm/wet
- **Hesperian** (3.7–3.0 Ga): outflow channels (catastrophic flood-carved), large-scale volcanism, lake remnants
- **Amazonian** (< 3.0 Ga): cold, dry, thin atmosphere

Mars Geological Map



Credit: Tanaka et al. (2014)

- Tharsis volcanic province dominates the western hemisphere
- Hellas, Argyre, Isidis: large impact basins
- Northern lowlands: relatively young, partly buried
- Map gives direct visual of stratigraphic ages

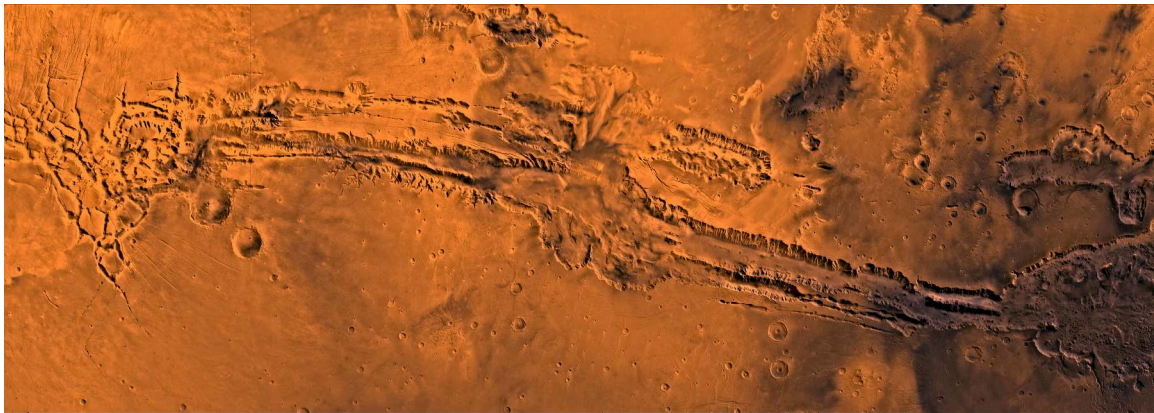


Olympus Mons, Mars Express. Credit: ESA/DLR/FU Berlin

Olympus Mons: The Tallest Volcano

- Summit ~ 21.2 km above the Mars areoid (its gravitational reference surface); ~ 21.9 km above the surrounding plains (Earth's tallest, Mauna Kea, is ~ 10 km from sea floor base)
- Diameter: 600 km (the size of Italy)
- Shield volcano with low slope angles
- Calderas at the summit: ~ 80 km across
- Flank scarps: ~ 6 km cliffs
- Stagnant-lid (rigid, non-recycling lithosphere) plumes can pile up to enormous heights without lateral plate motion to spread the load

Tharsis as a whole is loaded enough to have plausibly reoriented Mars' rotation axis ("true polar wander").



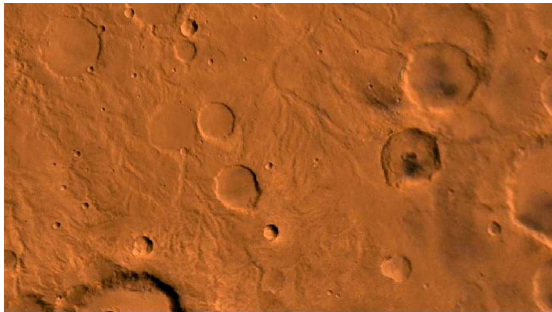
Valles Marineris, Viking mosaic. Credit: NASA/JPL-Caltech

Valles Marineris: A Tectonic Canyon

- 4000 km long, 200 km wide, up to 7 km deep
- As wide as the United States; as deep as four Grand Canyons stacked
- Origin: extensional rifting linked to Tharsis loading + uplift
- Outflow channels exit the eastern end into chaotic terrain (jumbled, collapsed surface blocks)
- Possibly lake-filled in the Hesperian; layered sediments observed

Valles Marineris is not an erosional canyon; it is a tectonic graben (a down-dropped, fault-bounded block) modified by collapse and subsequent fluvial activity.

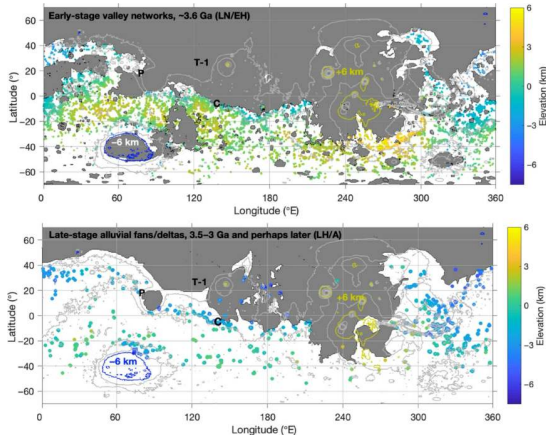
Past Water: Valley Networks



Credit: NASA/JPL/USGS

- Dendritic drainage networks on Noachian terrain
- Confluences, meanders, tributaries: liquid-water erosion
- Drainage density well below Earth's; densest networks in Earth's range (Hynek et al. 2010)
- Active for > 100 Myr in the late Noachian

Mapping the Wet Era



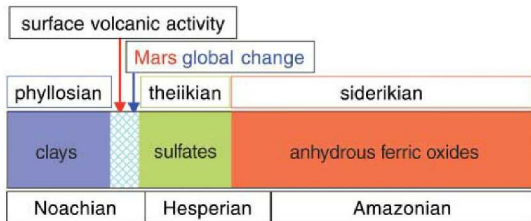
Credit: Kite et al. (2022), Fig. 1

- **Top:** Late Noachian / Early Hesperian valley networks (~ 3.6 Ga and older), on high ground
- **Bottom:** Late Hesperian / Amazonian fans and deltas (3.5–3 Ga), lower and banded at mid-latitudes
- Colour is elevation, -6 to +6 km; grey is masked out (detection unreliable)
- Control passes from **elevation** to **latitude**: the key constraint on the greenhouse change

Phyllosian, Theiikian, Siderikian: Mineralogy Tells Time

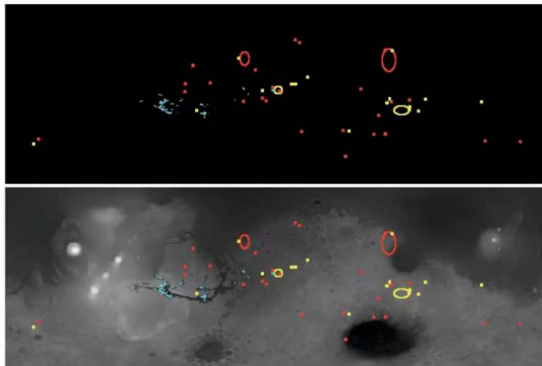
OMEGA (Mars Express)
mineralogy framework:

- **Phyllosian** (earliest): clays, neutral-pH water
- **Theiikian**: sulfates, acidic water
- **Siderikian** (from ~ 3.5 Ga): anhydrous ferric oxides, oxidising, dry
- Mineralogical eras, not chronostratigraphic: the boundaries differ from the Noachian / Hesperian / Amazonian ones



Credit: Bibring et al. (2006), Fig. 5

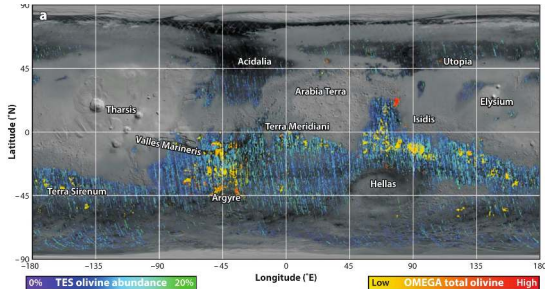
Phyllosilicate and Sulphate Distributions



Credit: Bibring et al. (2006), Fig. 3

- Red: phyllosilicates (clays)
- Blue: sulfates
- Yellow: other hydrated minerals
- Clays cluster in the southern Noachian highlands (neutral-pH water, phyllosian)
- Sulfates concentrate at lower latitudes (acidic evaporative, theiikian)

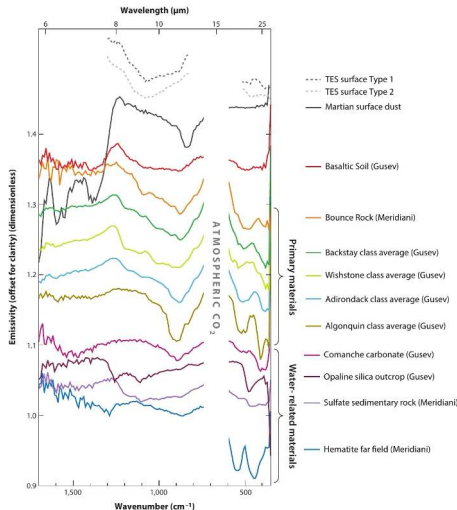
Olivine: The Aqueous Alteration Boundary



Credit: Ehlmann & Edwards (2014)

- Olivine is unstable in liquid water (alters to serpentine + clays)
- Detection of fresh olivine = surfaces never sustainedly wet
- Olivine common across the equatorial region: most of Mars never had a long surface ocean

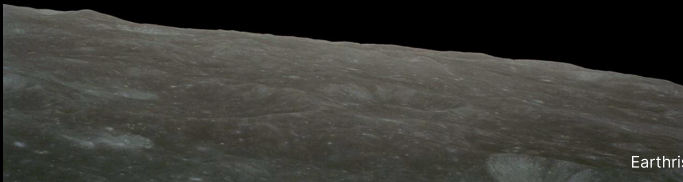
CRISM Spectra: How We See the Minerals



Credit: Ehlmann & Edwards (2014)

- Mid-IR emissivity spectra ($\sim 6\text{--}25\ \mu\text{m}$, wavenumber $1500\text{--}500\ \text{cm}^{-1}$)
- Atmospheric CO_2 band, basaltic minerals, sulfates, hematite endmembers
- Diversity of spectral classes records a wide range of aqueous and igneous environments
- Complementary near-IR CRISM spectra (not shown) resolve $1.4, 1.9, 2.2, 2.4\ \mu\text{m}$ hydration features

Break



6. Blackboard Derivation: Jeans Escape Flux



Setup: The Exobase

Goal: derive the rate at which thermal energy alone allows molecules to escape from a planetary atmosphere; apply it to Mars.

The exobase. Altitude where the mean free path equals the atmospheric scale height. Above it, molecules move ballistically.

Most-probable speed in Maxwell-Boltzmann distribution:

$$v_{\text{th}} = \sqrt{2k_{\text{B}}T/m}$$

Escape velocity at the exobase:

$$v_{\text{esc}} = \sqrt{2GM/r_{\text{exo}}}$$

Step 1: The Escape Parameter

Define dimensionless ratio:

$$\lambda \equiv \left(\frac{v_{\text{esc}}}{v_{\text{th}}} \right)^2 = \frac{G M m}{k_{\text{B}} T r_{\text{exo}}}$$

- $\lambda \gg 1$: molecules gravitationally bound; escape rare
- $\lambda \lesssim 1$: thermal energy comparable to escape energy; rapid loss
- Depends strongly on molecular mass m

Step 2: Maxwell-Boltzmann Distribution

At the exobase, gas in local thermal equilibrium:

$$f(v) dv = n \left(\frac{m}{2\pi k_B T} \right)^{3/2} 4\pi v^2 \exp\left(-\frac{mv^2}{2k_B T}\right) dv$$

Number flux of upward-moving molecules with $v_z > 0$:

$$\Phi = \int_{\text{up}} v_z f(\mathbf{v}) d^3v$$

We need v_z such that the speed exceeds v_{esc} .

Step 3: The Jeans Formula

Integrating over the upper hemisphere of velocity space, with the speed cutoff $v \geq v_{\text{esc}}$:

$$\Phi_J = n \langle v \rangle \frac{1}{4} (1 + \lambda) e^{-\lambda}$$

where $\langle v \rangle = \sqrt{8k_B T / (\pi m)}$ is the mean speed.

Substituting and simplifying:

$$\Phi_J = n \sqrt{\frac{k_B T}{2\pi m}} (1 + \lambda) e^{-\lambda}$$

This is the Jeans escape flux (Jeans 1925).

Step 4: Checking the Limits

- $\lambda \rightarrow 0$: $(1 + \lambda)e^{-\lambda} \rightarrow 1$, formula reduces to standard kinetic-theory effusion flux $n\sqrt{k_B T / (2\pi m)}$
- $\lambda \gg 1$: exponential decay dominates; flux $\propto \lambda e^{-\lambda}$
- Steepness: a change in λ from 5 to 10 reduces escape by a factor of ~ 80

Jeans escape is exponentially sensitive to λ : small changes in temperature or molecular mass change the loss rate by orders of magnitude.

Step 5: Apply to Mars

$M = 6.4 \times 10^{23} \text{ kg}$, $r_{\text{exo}} \approx 3.6 \times 10^6 \text{ m}$, $T_{\text{exo}} \approx 270 \text{ K}$.

Escape velocity at the exobase: $v_{\text{esc}} \approx 4.9 \text{ km/s}$

Species	λ	$e^{-\lambda}$
H	5.3	5×10^{-3}
H ₂	10.6	$2-3 \times 10^{-5}$
He	21.2	$\sim 10^{-9}$
O	84.8	10^{-37}
CO ₂	230	10^{-100}

Mars cannot lose CO₂ thermally. Only H and H₂ escape via Jeans; heavier species need non-thermal channels.

Why Jeans Escape is the Foundation

- Underpins atmospheric escape theory in general
- Always present at any temperature above absolute zero
- Sets the lower bound on escape rate
- Independent of magnetic field, ionisation state, or solar wind

Other mechanisms (photochemical, sputtering, ion pickup, hydrodynamic) operate **in addition** to Jeans, with different mass selectivities and intensities.



7. Mars Atmosphere and Climate Loss

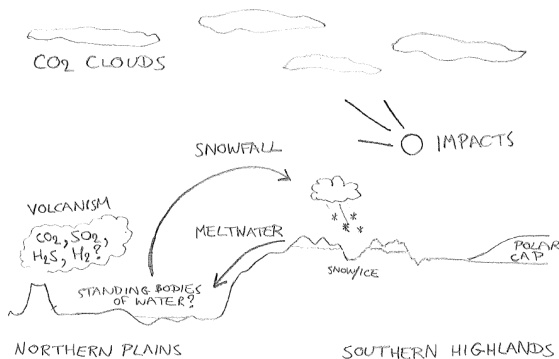
Dendritic channel networks in the cratered Martian highlands, imaged by the Viking Orbiters. Credit: NASA/JPL/USGS

The Early Mars Climate Puzzle

- Geological evidence: liquid water flowed in the late Noachian
- Solar luminosity at 4 Ga was ~ 75% of today
- Pure CO₂ greenhouse cannot warm Mars above freezing at 1.5 AU
- Required: extra greenhouse gas (CH₄, H₂ collision-induced absorption)

Active research: episodic warming + impacts + volcanism + reducing atmospheres in interplay.

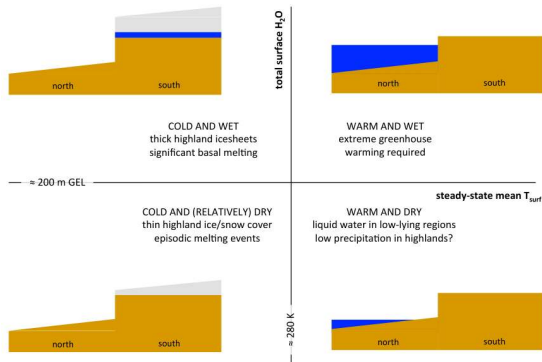
Reducing-Atmosphere Greenhouse



Credit: Wordsworth (2016); CIA: Wordsworth (2017)

- Icy-highlands picture: snow accumulates on the elevated southern terrain
- Episodic warming yields meltwater into the northern lowlands
- CO₂ clouds; volcanic outgassing of CO₂, SO₂, H₂S, H₂
- H₂-CO₂ collision-induced absorption (Wordsworth 2017) is the supporting greenhouse that can keep early Mars above freezing at 1.5 AU

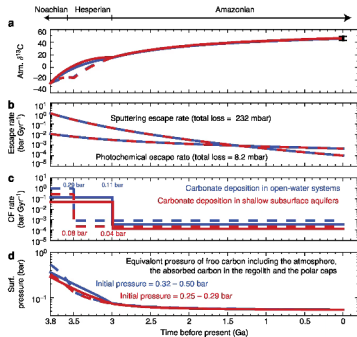
Phase Diagram: Wet vs Dry Mars



Credit: Wordsworth (2016), Fig. 7

- Steady-state mean T_{surf} (horizontal, ~ 280 K divider) vs. surface H₂O inventory (vertical, ~ 200 m GEL divider)
- Quadrant classification (warm/cold × wet/dry) of long-term states
- Cross-sections run lowlands (left) to highlands (right): rock orange, ice grey, water blue
- Geomorphology favours cold and relatively dry, with episodic melting

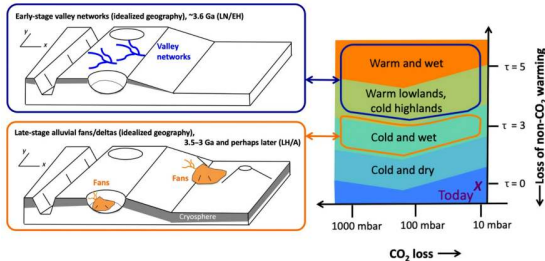
The Carbon Inventory



Credit: Hu et al. (2015)

- Coupled atmosphere-surface-interior carbon model
- Initial $\text{CO}_2 \sim 0.25\text{--}0.50$ bar (panel d; red 0.32–0.50, blue 0.25–0.29)
- Reconstructed Late Noachian / Early Hesperian total carbon $\sim 0.5\text{--}1.8$ bar
- Loss split between escape, carbonate burial, polar caps
- Modern 6 mbar: most of the original atmosphere is gone

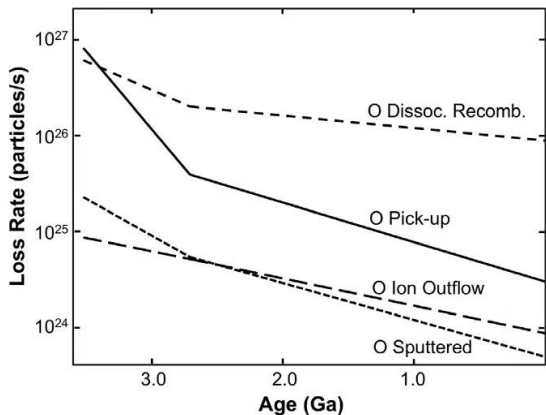
What Drove the Wet-to-Dry Transition?



Credit: Kite et al. (2022), Fig. 6

- **Left:** the two eras; valleys cut the highlands, later fans lie on lower ground above the cryosphere
- **Right:** climate state vs. CO₂ loss (1000 → 10 mbar) and lost non-CO₂ warming ($\tau = 5 \rightarrow 0$)
- Blue outline: early era. Orange: late era. Between them, cooling of ≥ 10 K, from $T \geq 268$ K to $T \sim 258$ K
- Driven mainly by **waning non-CO₂ forcing**, not CO₂ loss

MAVEN: Measuring Atmospheric Loss



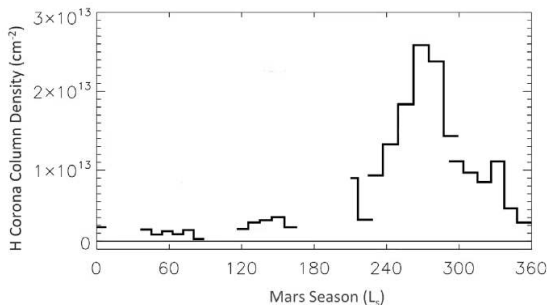
Credit: Jakosky et al. (2018)

Figure: **oxygen** escape channels vs. time, extrapolated from MAVEN to ~ 3.5 Ga:

- **Dissociative recombination** (dominant O channel today)
- **Pick-up ions** (steepest decline over time)
- **Ion outflow**
- **Sputtering** by solar-wind ions

Present-day total escape $\sim 2 \text{ kg s}^{-1}$; H and CO_2^+ channels (not shown here) contribute additionally.

Seasonal H Corona Variation



Credit: Jakosky et al. (2018)

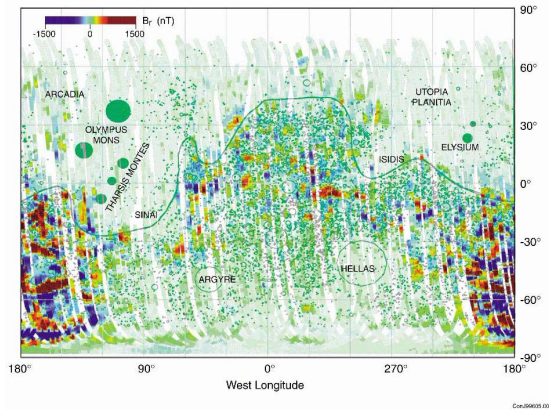
- H corona column density vs. L_s (solar longitude, a seasonal marker) over a Mars year
- Peak near $L_s \sim 270^\circ$: perihelion warming lifts H_2O to photolysis altitudes
- Order-of-magnitude seasonal variation; proxy for the escape rate
- Loss is Jeans + non-thermal; over 4 Gyr implies a much wetter early Mars



8. Mars Magnetism and Mission History

Early Mars from Mariner 4 (1965). Credit: NASA/JPL

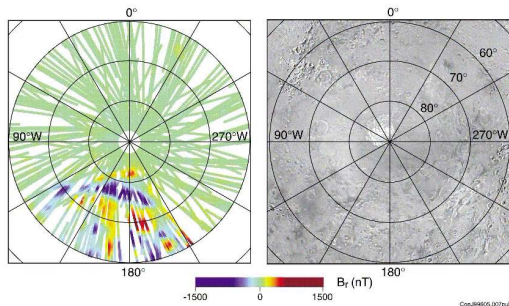
Mars Crustal Magnetism



Credit: Acuña et al. (1999)

- No global field today
- Strong remnant magnetisation in the southern highlands
- Implies a vigorous early dynamo, then shutoff
- Northern lowlands: weak (impact-erased or younger crust)

Lineated Magnetic Anomalies



Credit: Acuña et al. (1999)

- Polar stereographic B_r ; companion equatorial map shows mid-latitude stripes (polarity reversals)
- Recorded by basaltic crust during cooling
- Pre-Borealis-impact crust preserves the early dynamo signature
- Dynamo cessation ~ 4.1 Ga (Acuña 1999); refined to ~ 3.7 Ga (Mittelholz 2020, Steele 2024)

Mars Mission History (Selected Highlights)

- **Mariner 4** (1965): first flyby; cratered "Moon-like" Mars
- **Viking 1&2** (1976): first landers, first life-detection experiments (ambiguous results)
- **MGS, Odyssey, MEX, MRO, Maven**: orbital era 1996–present
- **Spirit, Opportunity, Curiosity, Perseverance**: rovers
- **InSight** (2018–2022): seismometer + heat-flow probe
- **Tianwen-1** (2021, China): orbiter + Zhurong rover
- **Mars Sample Return**: planned 2030s

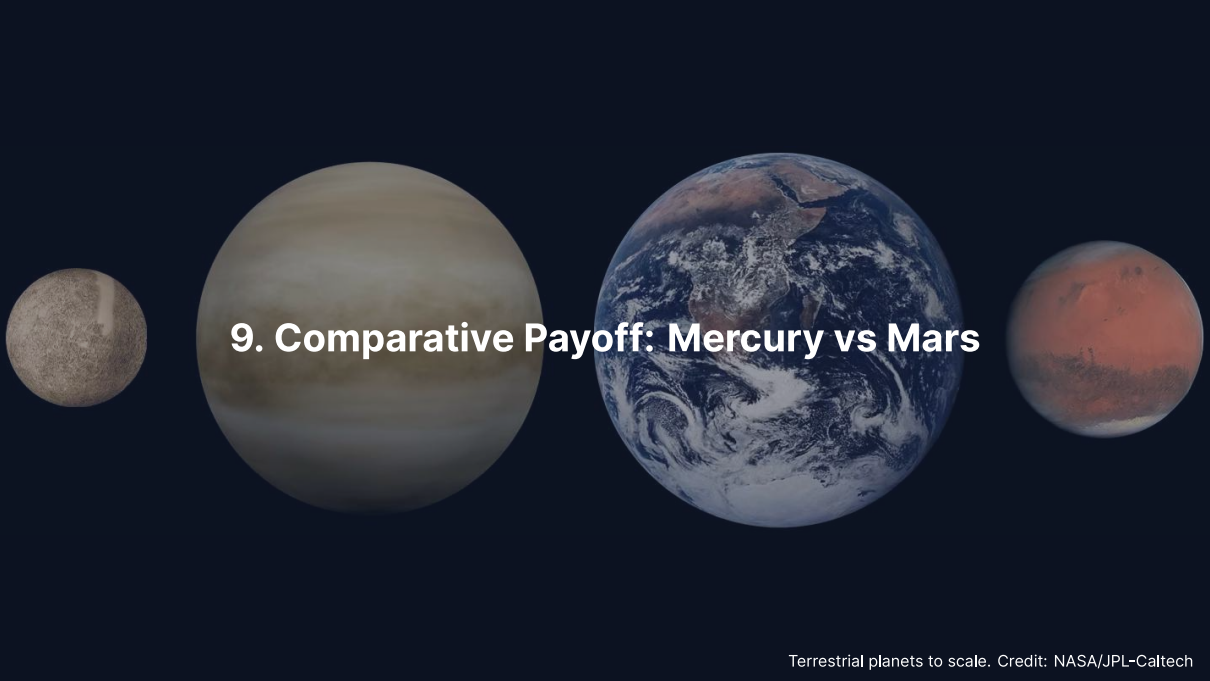
Mars Sample Return: The Next Decade

Perseverance (since 2021) is caching samples from Jezero Crater, an ancient lake bed.

- Cores: igneous, sedimentary, organic-bearing rocks
- ESA + NASA architecture under revision (cost overruns; IRB 2023)
- Sample return earliest in the early 2030s
- Will allow precise radiometric dating + biosignature analysis

Sample return is the difference between “consistent with biological origin” and “biological origin proven beyond reasonable doubt”.

Source: NASA/ESA MSR Independent Review Board (2023)

A horizontal arrangement of four celestial bodies against a dark blue background. From left to right: a small, grey, cratered sphere (Mercury); a large, yellowish-brown sphere with horizontal cloud bands (Venus); a large sphere showing blue oceans, white clouds, and brown landmasses (Earth); and a medium-sized, reddish-orange sphere with darker patches (Mars). The text '9. Comparative Payoff: Mercury vs Mars' is centered over the Venus and Earth spheres.

9. Comparative Payoff: Mercury vs Mars

Mercury and Mars as Opposite Limits

Property	Mercury	Mars
Distance from Sun	0.39 AU	1.52 AU
Mass ($M_{\oplus}$)	0.055	0.107
Core fraction	~ 74%	~ 25%
Mantle thickness	~ 400 km	~ 1560 km
Surface volatiles	cold trap ice only	once-flowing water
Magnetic field today	weak global	none global
Atmosphere	exosphere	6 mbar CO ₂
Tectonics	shrinking, lobate scarps	stagnant lid

Mercury sits at the maximum-iron, no-volatile end. Mars sits at the maximum-volatile, low-iron end. Earth and Venus fall in between.

Size and Distance Set the Trajectory

- **Small bodies cool fast:** surface-area-to-volume ratio $\propto 1/R$
- Mars cools faster than Earth $\Rightarrow$ early dynamo shutoff
- Mercury sustains a weak dynamo because light elements (S, Si) keep the outer core liquid; a thin convecting shell + thermal stratification produce the offset, weak field
- **Distance from the Sun** sets atmospheric escape and irradiation history

Combined: small + warm $\Rightarrow$ Mercury (volatile-poor, iron-dominant). Small + cold $\Rightarrow$ Mars (water-rich early, escapes to space later).

The Timing Problem: Dynamo Lifetimes

- Earth: dynamo active for ≥ 3.5 Gyr, still going
- Mars: dynamo active in the first ~ 500 – 800 Myr; cessation ~ 4.1 Ga (Acuña 1999), possibly as late as ~ 3.7 Ga (Mittelholz 2020, Steele 2024)
- Mercury: dynamo weak but still active
- Venus: dynamo unclear, possibly dead

Lesson: dynamo persistence depends on (a) core size, (b) cooling rate of the mantle, (c) chemical buoyancy from inner-core nucleation. Mars failed (b); Mercury preserves (a); Earth has all three.

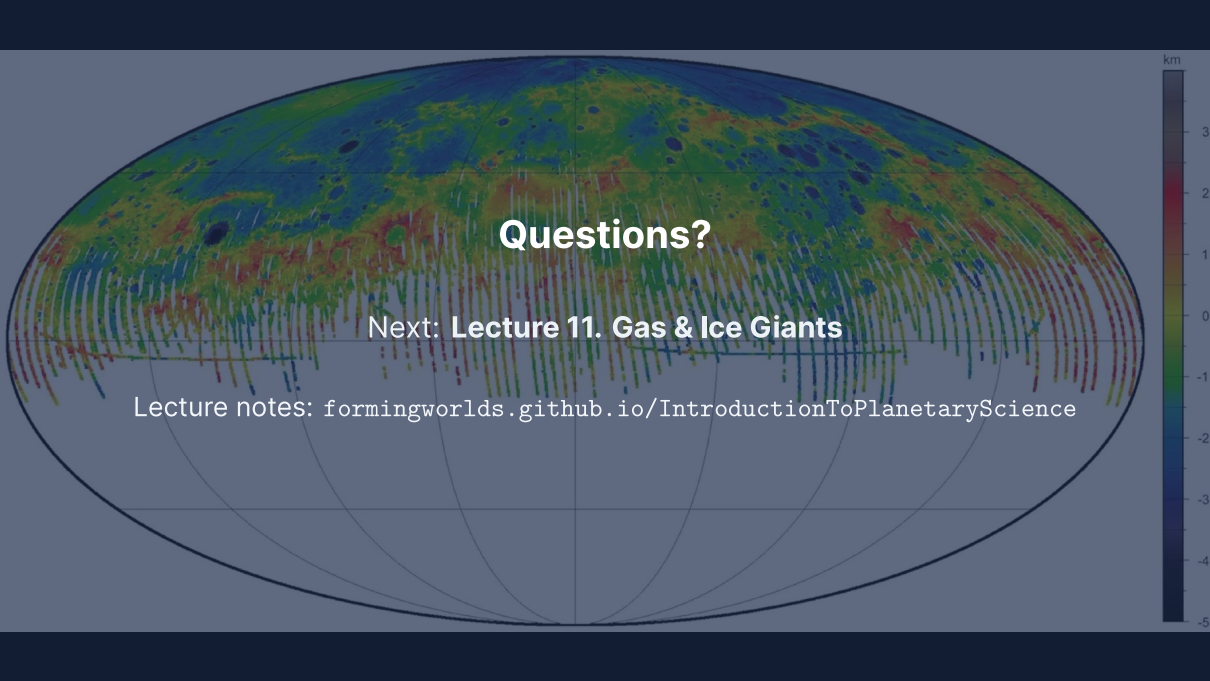
Comparative View: What Makes a Rocky Planet Habitable?

- Stellar flux inside the runaway-greenhouse limit (rules out Venus)
- Long-lasting atmosphere (rules out Mercury, late Mars)
- Liquid water reservoir at the surface (rules out Mercury entirely)
- Internal heat sufficient for tectonic recycling (rules out Mars after ~ 1 Ga)
- Magnetic shield against solar-wind erosion (helps but not strictly required)

Habitability is a multi-dimensional constraint. Each of Mercury, Venus, Mars fails on different axes; Earth happens to satisfy all of them simultaneously.

Summary

1. Mercury: 3:2 spin-orbit resonance, ~ 74% iron core, weak offset dynamo, polar ice in cold traps
2. Mars: half Earth's radius, one tenth its mass, lost its dynamo at ~ 4.1 Ga and most of its atmosphere thereafter
3. Jeans escape: $\Phi_j \propto (1 + \lambda)e^{-\lambda}$, mass-selective; H and H₂ escape Mars, CO₂ does not
4. Non-thermal escape (photochemical, ion pickup, sputtering) drives the heavy-species loss
5. InSight delivered the first marsquake catalogue and confirmed a liquid Mars core
6. Mercury and Mars sit at opposite limits of terrestrial-planet evolution: maximum core fraction vs. maximum atmospheric volatile loss
7. BepiColombo (2026) and MMX (2026) will revolutionise our view of both planets in the next few years



Questions?

Next: **Lecture 11. Gas & Ice Giants**

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience